

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

NAGAPATTINAM – 611002

*(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade | Accredited
 by NBA T1 (B.E. – CSE, CIVIL, ECE, EEE, MECH & B.Tech – IT,) |
 Approved by AICTE, New Delhi)*



MASTER OF COMPUTER APPLICATIONS R - 2024

FIRST YEAR

CURRICULUM AND SYLLABUS FOR SECOND SEMESTER

Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CIA	ES	Total	
Theory Course									
2402CA201	Full Stack Web Development	3	-	-	3	40	60	100	PC
2402CA202	Advanced Database Technologies	3	-	-	3	40	60	100	PC
2402CA203	Mobile Application Development	2	-	2	3	50	50	100	PC
2402CA204	Cloud Computing	3	-	-	3	40	60	100	PC
2402CA205	Software Testing	3	-	-	3	40	60	100	PC
	Professional Elective–I	3	-	-	3	40	60	100	PE
	Audit Course II*	2	-	-	-	100	0	100	AC
Laboratory Course									
2402CA206	Full Stack Web Development Laboratory	-	-	4	2	60	40	100	PC
2402CA207	Advanced Database Technologies Laboratory	-	-	4	2	60	40	100	PC
2404CA208	Professional Development Course II	-	-	2	1	100	-	100	EEC
Total		19	0	12	23	570	430	1000	

L-Lecture | T –Tutorial | P- Practical | CA – Continuous Assessment | ES – End Semester

2402CA201

FULL STACK WEB DEVELOPMENT

L	T	P	C
3	0	0	3

Prerequisite Web Development

COURSE OBJECTIVES

1. To understand the fundamentals of web programming and client side scripting.
2. To learn server side development using NodeJS.
3. To understand API development with Express Framework.
4. To understand and architect databases using NoSQL and SQL databases.
5. To learn the advanced client side scripting and React JS framework

COURSE OUTCOMES

- At the end of the course, students will be able to
- CO1 Develop client side scripting using HTML, CSS and JS..
- CO2 Develop server side application using NodeJS
- CO3 Create NoSQL databases with MongoDB
- CO4 Develop web application using ReactJS.
- CO5 Implement a full-stack Single Page Application using React, NodeJS and MongoDB and deploy on Cloud.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	3	3	-	-	-	3	3	-
CO2	3	3	3	3	-	-	-	3	-	3
CO3	3	3	3	3	-	-	-	3	-	3
CO4	3	3	3	3	-	-	-	3	3	-
CO5	3	3	3	3	-	-	-	3	3	3
Avg	3	3	3	3	-	-	-	3	1.8	1.8

UNIT I INTRODUCTION TO CSS and JAVASCRIPT

9 Hours

Introduction to Web: Server - Client - Communication Protocol (HTTP) – Structure of HTML Documents – Basic Markup tags – Working with Text and Images with CSS– CSS Selectors – CSS Flexbox - JavaScript: Data Types and Variables - Functions - Events – AJAX: GET and POST

UNIT II SERVER SIDE PROGRAMMING WITH NODE JS

9 Hours

Introduction to Web Servers – Javascript in the Desktop with NodeJS – NPM – Serving files with the http module – Introduction to the Express framework – Server-side rendering with Templating Engines – Static Files - async/await - Fetching JSON from Express

UNIT III ADVANCED NODE JS AND DATABASE

9 Hours

Introduction to NoSQL databases – MongoDB system overview - Basic querying with MongoDB shell – Request body parsing in Express – NodeJS MongoDB connection – Adding and retrieving data to MongoDB from NodeJS – Handling SQL databases from NodeJS – Handling Cookies in NodeJS – Handling User Authentication with NodeJS

UNIT IV ADVANCED CLIENT SIDE PROGRAMMING

9 Hours

React JS: ReactDOM - JSX - Components - Properties – Fetch API - State and Lifecycle - JS LocalStorage - Events - Lifting State Up - Composition and Inheritance

UNIT V APP IMPLEMENTATION IN CLOUD

9 Hours

Cloud providers Overview – Virtual Private Cloud – Scaling (Horizontal and Vertical) – Virtual Machines, Ethernet and Switches – Docker Container – Kubernetes

TOTAL : 45 PERIODS

REFERENCES:

1. David Flanagan, "Java Script: The Definitive Guide", O'Reilly Media, Inc, 7th Edition, 2020
2. Matt Frisbie, "Professional JavaScript for Web Developers", Wiley Publishing, Inc, 4th Edition, ISBN: 978-1-119-36656-0, 2019
3. Alex Banks, Eve Porcello, "Learning React", O'Reilly Media, Inc, 2nd Edition, 2020
4. Marc Wandschneider, "Learning Node", Addison-Wesley Professional, 2nd Edition, 2016
5. Joe Bada, Kelsey Hightower, Brendan Burns, Up and Media, 1st edition, 2020
6. Paul Zikopoulos, Christopher Bienko, Chris Backer, Chris Konarski, Sai Vennam, "Without Cloud Compromise", O'Reilly Media, 1st edition, 2021

2402CA202

ADVANCED DATABASES TECHNOLOGIES

L	T	P	C
3	0	0	3

Prerequisite

1. Basic Data Structures
2. Database management system

COURSE OBJECTIVES

1. To understand the working principles and query processing of distributed databases.
2. To understand the basics of spatial, temporal and mobile databases and their applications
3. To distinguish the different types of NoSQL databases.
4. To understand the basics of XML and create well-formed and valid XML documents.
5. To gain knowledge about information retrieval and web search.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Design a distributed database system and execute distributed queries.
 - CO2 Manage Spatial and Temporal Database systems and implement it in corresponding applications
 - CO3 Use NoSQL database systems and manipulate the data associated with it.
 - CO4 Design XML database systems and validate with XML schema.
 - CO5 Apply knowledge of information retrieval concepts on web databases.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	3	3	2	1	–	2	3	3
CO2	2	2	3	2	–	1	–	2	2	3
CO3	3	2	3	3	1	–	–	2	3	3
CO4	3	2	2	3	1	–	–	2	3	2
CO5	2	3	2	2	2	–	–	3	2	3
Avg	2.6	2.2	2.6	2.6	1.5	1	-	2.2	2.6	2.8

UNIT I DISTRIBUTED DATABASES

Distributed Systems – Introduction – Architecture – Distributed Database Concepts – Distributed Data Storage – Distributed Transactions – Commit Protocols – Concurrency Control – Distributed Query Processing

9 Hours

UNIT II SPATIAL AND TEMPORAL DATABASES

Active Databases Model – Design and Implementation Issues - Temporal Databases - Temporal Querying - Spatial Databases: Spatial Data Types, Spatial Operators and Queries – Spatial Indexing and Mining – Applications – Mobile Databases: Location and Handoff Management, Mobile Transaction Models – Deductive Databases - Multimedia Databases.

9 Hours

UNIT III NOSQL DATABASES

NoSQL – CAP Theorem – Sharding - Document based – MongoDB Operation: Insert, Update, Delete, Query, Indexing, Application, Replication, Sharding–Cassandra: Data Model, Key Space, Table Operations, CRUD Operations, CQL Types – HIVE: Data types, Database Operations, Partitioning – Hive QL – Orient DB Graph database – Orient DB Features

9 Hours

UNIT IV XML DATABASES

Structured, Semi structured, and Unstructured Data – XML Hierarchical Data Model – XML Documents – Document Type Definition – XML Schema – XML Documents and Databases – XML Querying – XPath – XQuery

9 Hours

UNIT V INTELLIGENT DATABASES

IR concepts – Retrieval Models – Queries in IR system – Text Preprocessing – Inverted Indexing – Evaluation Measures – Web Search and Analytics – Current trends.

9 Hours

TOTAL : 45 PERIODS

REFERENCES:

1. Abraham Silberschatz, Henry F Korth, S. Sudharshan, "Database System Concepts". Seventh Edition, McGraw Hill, 2019.
2. R. Elmasri, S.B. Navathe, "Fundamentals of Database Systems", Seventh Edition, Pearson Education/Addison Wesley, 2017.
3. Guy Harrison, "Next Generation Databases, NoSQL, NewSQL and Big Data", First Edition, Apress publishers, 2015
4. Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques", Third Edition, Morgan Kaufmann, 2012.
5. Brad Dayley, "Teach Yourself NoSQL with MongoDB in 24 Hours", Sams Publishing, First Edition, 2014.
6. C. J. Date, A. Kannan, S. Swamynathan, "An Introduction to Database Systems", Eighth Edition, Pearson Education, 2006

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Approved in 11th Academic Council Meeting held on 09.01.2024

2402CA203

MOBILE APPLICATION DEVELOPMENT

L	T	P	C
2	0	2	3

Prerequisite Programming in C / Java
 Web Technology

COURSE OBJECTIVES

1. To understand the need and characteristics of mobile applications.
2. To design the right user interface for mobile applications.
3. To understand the design issues in the development of mobile applications.
4. To understand the development procedure for mobile applications.
5. To develop mobile applications using various tools and platforms

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1: Apply mobile application development frameworks and tools to build basic mobile apps.
 CO2: Design effective user interfaces using UI components and event handling.
 CO3: Develop mobile apps with efficient memory, resource, and database management.
 CO4: Build and deploy mobile apps on Android and iOS using native or cross-platform tools.
 CO5: Integrate web and cloud services like Firebase, AWS, and Google Cloud into mobile apps.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	3	1	-	-	2	3	2
CO2	2	2	3	3	2	-	-	1	3	2
CO3	3	3	3	3	1	-	-	2	3	3
CO4	3	3	3	3	2	1	-	2	3	3
CO5	3	2	3	3	2	1	-	3	3	3
Avg	2.8	2.4	2.8	3	1.6	1	-	2	3	2.6

UNIT I INTRODUCTION

12 Hours

Mobile applications -Mobile Application Model – Characteristics and Benefits- Comparing Native vs. Hybrid Applications- Infrastructure and Managing Resources – Mobile Software Engineering- The Mobile Application Development Lifecycle -Mobile Device Profiles – Frameworks and Tools - -The Mobile Application Front-End-The Mobile Application Back-End

Lab Component:

1. Install and configure java development kit (JDK), android studio and android SDK
2. Implement mobile applications using frameworks
3. Develop an application that uses GUI components, fonts and colours.

UNIT II USER INTERFACE

12 Hours

Generic UI Development - UI Components - Event Handling - activity and its lifecycle – Material Design: new themes, new widgets, Cardlayouts – Backward compatibility – v7 appcompat library – Intent object, intent filters, adding categories – Menus – fragment and its lifecycle- Multimodal and Multichannel UI –Gesture Based UI – Screen Elements and Layouts –Drawing and Working with Animation. Playing Audio and Video, Recording Audio and Video, U Voice XML.

Lab Component:

1. Implement mobile applications using UI toolkits.
2. Design an application that uses Layout Managers and event listeners.
3. Build a simple native calculator application to do simple arithmetic operations.
4. Create animations and graphical primitives in Android environment.

UNIT III APPLICATION DESIGN

12 Hours

Memory Management – Design Patterns for Limited Memory – Workflow for Application development – Java API – Dynamic Linking – Plugins and rule of thumb for using DLLs – Multithreading in Java - Concurrency and Resource Management-Architecture – SDK Tools – Languages for Android - XML – Java / Kotlin - UI Widgets – Layouts-Overview of Application Components - Android Intents, Types - SQLite Database – CRUD

Lab Component:

1. Design a mobile application that is aware of the resource constraints of mobile devices.
2. Design an application that uses Dynamic Linking
3. Develop an application that makes use of SQL Lite mobile database.
4. Implement an application that uses Multi-threading

UNIT IV MOBILE OS

12 Hours

Mobile OS: Android, iOS – Android Application Architecture – Understanding the anatomy of a mobile application - Android basic components –Intents and Services – Storing and Retrieving data – Packaging and Deployment – Security and Hacking-Android architecture –ART(Android Runtime) – ADB(Android Debug Bridge) – Application framework basics-Java &Kotlin for Android, Swift & Objective-C for iOS, Basics of React Native, Native Components, JSX, State, Props.

Lab Component:

1. Develop an application that makes use of mobile database
2. Implement an android application that writes data into the SD card.
3. Write a mobile application that makes use of RSS feed

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UNIT V

APP IMPLEMENTATION IN CLOUD

12 Hours

Communication via the Web — Notification and Alarms — Graphics and Multimedia: Layer Animation, Event handling and Graphics services — Telephony — Location based services- Cloud Database Connectivity: Firebase, AWS, Google Cloud-Consuming Web Services Using HTTP- Consuming JSON Services-Popular Cross-platform App Development Frameworks, Flutter, Xamarin, React-Native, Basics of React Native, Native Components, JSX, State, Props.

Lab Component:

1. Develop a web based mobile application that accesses internet and location data.
2. Develop an android application using telephony to send SMS.
3. Implement simple gaming application using open-source tools like flutter or Unity

TOTAL : 60 PERIODS

REFERENCES:

1. Reto Meier, Ian Lake, “Professional Android”, 4th Edition, Wrox, 2018.
2. Zigurd Mednieks, Laird Dornin, G. Blake Meike, Masumi Nakamura, “Programming Android”, O’Reilly, 2nd Edition, 2012.
3. Alasdair Allan, “Learning iOS Programming”, O’Reilly, Third Edition, 2013.
4. Bill Phillips, Chris Stewart, Brian Hardy, and Kristin Marsicano, Android Programming: The Big Nerd Ranch Guide, 4th edition, 2019.
5. Christian Keur, Aaron Hillegass, iOS Programming: The Big Nerd Ranch Guide, 6th Edition, O’Reilly, 2016.
6. Barry Burd, “Android Application Development All-In-One for Dummies”, 3rd Edition, 2021

2402CA204	CLOUD COMPUTING	L	T	P	C
Prerequisite	1.Computer Networks 2.Operating Systems	3	0	0	3

COURSE OBJECTIVES

1. To understand the basic concepts of Distributed systems.
2. To learn about the current trend and basics of Cloud computing.
3. To be familiar with various Cloud concepts.
4. To expose with the Server, Network and storage virtualization
5. To be aware of Microservices and DevOps

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Apply the concepts of Distributed systems in Cloud Environment.
- CO2 Apply basic cloud computing concepts and features to develop simple cloud applications.
- CO3 Implement basic cloud-enabled solutions by integrating cloud infrastructure, service models, and IoT components for a given problem statement.
- CO4 Apply cloud enabling technologies such as virtualization, SOA, and IoT for effective deployment.
- CO5 Design, develop, and deploy cloud-based applications using micro services architecture and DevOps tools

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	-	-	-	2	3	2
CO2	3	2	2	3	-	-	-	2	3	3
CO3	2	3	3	3	1	1	1	2	3	3
CO4	2	2	3	3	1	-	1	2	3	3
CO5	2	3	3	3	2	2	1	3	3	3
Avg	2.4	2.4	2.6	2.8	2	1.5	1	2.2	3	2.8

UNIT I **DISTRIBUTED SYSTEMS** 9 Hours

Introduction to Distributed Systems – Characterization of Distributed Systems – Distributed Architectural Models – Remote Invocation – Request-Reply Protocols – Remote Procedure Call – Remote Method Invocation – Group Communication – Coordination in Group Communication – Ordered Multicast – Time Ordering – Physical Clock Synchronization – Logical Time and Logical Clocks.

UNIT II **BASICS OF CLOUD COMPUTING** 9 Hours

Cloud Computing Basics – Desired features of Cloud Computing – Elasticity in Cloud – On demand provisioning - Applications – Benefits – Cloud Components: Clients, Datacenters & Distributed Servers – Characterization of Distributed Systems – Distributed Architectural Models - Principles of Parallel and Distributed computing - Applications of Cloud computing – Benefits – Cloud services – Open source Cloud Software: Eucalyptus, Open Nebula, Open stack, Aneka, Cloudsim

UNIT III **CLOUD INFRASTRUCTURE** 9 Hours

Cloud Architecture and Design – Architectural design challenges – Technologies for Network- Based system - NIST Cloud computing Reference Architecture – Public, Private and Hybrid clouds – Cloud Models : IaaS, PaaS and SaaS – Cloud storage providers - Enabling Technologies for the Internet of Things – Innovative Applications of the Internet of Things

UNIT IV **CLOUD ENABLING TECHNOLOGIES** 9 Hours

Service Oriented Architecture – Web Services – Basics of Virtualization – Emulation – Types of Virtualization – Implementation levels of Virtualization – Virtualization structures – Tools & Mechanisms – Virtualization of CPU, Memory & I/O Devices – Desktop Virtualization – Server Virtualization – Google App Engine - AWS - Federation in the Cloud.

UNIT V **MICROSERVICES AND DEVOPS** 9 Hours

Defining Microservices - Emergence of Microservice Architecture – Design patterns of Microservices – The Mini web service architecture – Microservice dependency tree – Challenges with Microservices - SOA vs Microservice – Microservice and API – Deploying and maintaining Microservices – Reason for having DevOps – Overview of DevOps – Core elements of DevOps –Life cycle of DevOps –Adoption of DevOps-- DevOps Tools – Build, Promotion and Deployment in DevOps.

TOTAL :45 PERIODS

REFERENCES:

1. Kai Hwang, Geoffrey C. Fox & Jack J. Dongarra, "Distributed and Cloud Computing, From Parallel Processing to the Internet of Things", Morgan Kaufmann Publishers, First Edition, 2012
3. Andrew S. Tanenbaum & Maarten Van Steen, "Systems - Principles Third Edition", Pearson, 2017.
4. Thomas Erl, Zaigham Mahood & Ricardo Puttini, "Computing, Concept, Technology & Prentice Hall, Second Edition, 2013.
5. Richard Rodger, "Tao of ISBN 9781617293146", Manning Publications, First Edition, December 2017
6. "On Microservices with Spring Boot and Spring Cloud: Build and deploy microservices using spring cloud, Istio and Packt Publishing Ltd, First Edition, September 2019.

2402CA205

SOFTWARE TESTING

L	T	P	C
3	0	0	3

Prerequisite Software Engineering Principles

COURSE OBJECTIVES

1. To introduce the basics and necessity of software testing.
2. To provide various testing techniques along with concepts of software bugs and its impact.
3. To develop and validate a test plan.
4. To build a testing team required.
5. To understand the need for and challenges in test automation and to develop testing scripts.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Understand the insight of software testing.
- CO2 Apply black box testing and white box testing for software applications
- CO3 Apply multiple levels of testing for software applications
- CO4 Understand the role of a tester as an individual and as a team member.
- CO5 Apply software testing for large projects using automated testing tools and Maintain documentation on testing.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	2	–	–	–	2	3	–
CO2	3	3	2	3	–	–	–	–	3	–
CO3	3	3	3	3	–	–	–	–	3	–
CO4	2	–	–	–	3	–	2	–	2	–
CO5	3	2	3	3	2	3	1	2	3	–
Avg	2.8	2.5	2.2	2.7	2.5	3	1.5	2	2.8	-

UNIT I TESTING PRINCIPLES AND AXIOMS 9 Hours
 Testing as a Process – Testing Axioms –Software Testing Principles – Origins and Cost of Defects – Defect Classes and Examples – Developer/Tester Support of Developing a Defect Repository – Defect Prevention Strategies

UNIT II BLACK BOX, WHITE BOX TESTING AND TEST ADEQUACY 9 Hours
 Test Case Design Strategies – Black Box Approach – Boundary Value Analysis – Equivalence Class Partitioning – State-Based Testing – User Documentation Testing – White Box Approach – Static Testing vs. Structural Testing – Code Functional Testing – Coverage and Control Flow Graphs – Covering Code Logic – Paths – Cyclomatic Complexity – Test Adequacy Criteria.

UNIT III LEVELS OF TESTING 9 Hours
 Unit Test – Planning – Designing the Unit Test Process – Running the Unit Tests and Recording Results – Integration Test Planning – Scenario Testing – Defect Bash Elimination System Testing – Acceptance Testing – Performance Testing – Regression Testing – Internationalization Testing – Ad-Hoc Testing – Alpha, Beta Tests.

UNIT IV TEST MANAGEMENT 9 Hours
 Organization Structures for Testing Teams – Testing Services – Test Planning Attachments – Locating Test Items – Test Management – Reporting Test Results – The Role of Three Groups in Test Planning and Policy Development – Introducing the Test Specialist – Skills Needed by a Test Specialist – Building a Testing Group.

UNIT V TEST AUTOMATION 9 Hours
 Skill Needed for Automation – Scope of Automation – Design and Architecture for Automation – Requirements for a Test Tool – Challenges in Automation – Test Metrics and Measurements – Project, Progress and Productivity Metrics – Maintenance of Documents During Testing.

TOTAL :45 PERIODS

REFERENCES:

1. Paul C. Jorgensen, “Software Testing: A Craftsman's Approach”, Fourth Edition, CRC Press, 2013.
2. Dorothy Graham, Mark Fewster, “Experiences of Test Automation: Case Studies of Software Test Automation”, Pearson Education, 2012
3. Glenford J. Myers, Tom Badgett, Corey Sandler, “The Art of Software Testing”, Third edition, John Wiley & Sons publication, 2012.
4. Srinivasan Desikan, Gopalaswamy Ramesh, “Software Testing - Principles and Practices”, Pearson Education, 2009.
- Boris Beizer, “Software testing techniques”, Dream Tech Press, 2009.
5. Mauro Pezze, Michal Young, “Software Testing and Analysis Process Principles and Techniques”, Wiley India, 2008.

2402CA206 FULL STACK WEB DEVELOPMENT LABORATORY L T P C
Prerequisite Web Technologies Basics 0 0 4 2

COURSE OBJECTIVES

1. To implement the client side of the web application using javascript
2. To understand Javascript on the desktop using NodeJS
3. To develop a web application using NodeJS and Express.
4. To implement a SPA using React.
5. To develop a full stack single page application using React, NodeJS, and a Database (MongoDB or SQL)

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Implement and deploy the client side of the web application.
- CO2 develop and deploy server side applications using NodeJS
- CO3 use Express framework in web development
- CO4 Implement and architect database systems in both NoSQL and SQL environments.
- CO5 Develop a full stack single page application using React, NodeJS, and a Database and deploy using containers.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	3	2	–	–	2	3	–
CO2	3	3	3	3	2	–	–	2	3	–
CO3	2	2	3	3	–	–	–	2	3	–
CO4	3	3	3	3	–	–	–	2	3	2
CO5	3	3	3	3	3	3	–	3	3	3
Avg	2.8	2.6	2.8	3	2.3	3	-	2.2	3	2.5

List of experiments:

1. Create a form and validate the contents of the form using JavaScript.
2. Get data using Fetch API from an open-source endpoint and display the contents in the form of a card.
3. Create a NodeJS server that serves static HTML and CSS files to the user without using Express.
4. Create a NodeJS server using Express that stores data from a form as a JSON file and displays it in another page. The redirect page should be prepared using Handlebars.
5. Create a NodeJS server using Express that creates, reads, updates and deletes students' details and stores them in MongoDB database. The information about the user should be obtained from a HTML form.
6. Create a NodeJS server that creates, reads, updates and deletes event details and stores them in a MySQL database. The information about the user should be obtained from a HTML form.
7. Create a counter using ReactJS
8. Create a To do application using ReactJS. Store the data to a JSON file using a simple NodeJS server and retrieve the information from the same during page reloads.
9. Create a simple Sign up and Login mechanism and authenticate the user using cookies. The user information can be stored in either MongoDB or MySQL and the server should be built using NodeJS and Express Framework.
10. Create and deploy a virtual machine using a virtual box that can be accessed from the host computer using SSH.
11. Create a docker container that will deploy a NodeJS ping server using the NodeJS image.

TOTAL :60 PERIODS

Software Requirements

- 1.NodeJS/Express JS, ReactJS, Docker, any IDE like NOTEPAD++/visual studio code/sublime text etc.,
- 2.MySQL, MongoDB

2402CA207

ADVANCED DATABASE TECHNOLOGY LABORATORY

L	T	P	C
0	0	4	2

Prerequisite	Data Management System
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COURSE OBJECTIVES

1. To understand the process of distributing tables across multiple systems
2. To understand the process of storing, retrieving spatial and temporal data
3. To understand the process of storing, retrieving objects in a database
4. To understand the process of storing and retrieving data from a XML Database
5. To use the open source database for building a mobile application

COURSE OUTCOMES

On completion of the course, the student will be able to:

- CO1 Design and implement advanced databases.
- CO2 Use big data frameworks and tools.
- CO3 Formulate complex queries using SQL.
- CO4 Create an XML document and perform Xquery.
- CO5 Query processing in Mobile databases using open source tools.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	3	2	1	-	-	1	3	2
CO2	2	3	2	3	1	-	-	2	3	3
CO3	3	3	3	2	-	-	-	1	3	3
CO4	2	2	3	3	-	-	-	1	3	2
CO5	2	3	2	3	1	-	-	2	2	3
Avg	2.4	2.6	2.6	2.6	1	-	-	1.4	2.8	2.6

LIST OF EXPERIMENTS:

- 1 NOSQL Exercises
 - a. MongoDB – CRUD operations, Indexing, Sharding
 - b. Cassandra: Table Operations, CRUD Operations, CQL Types
- 2 MySQL Database Creation, Table Creation, Query
- 3 MySQL Replication – Distributed Databases
- 4 Spatial data storage and retrieval in MySQL
- 5 Temporal data storage and retrieval in MySQL
- 6 Object storage and retrieval in MySQL
- 7 XML Databases , XML table creation, XQuery FLWOR expression
- 8 Mobile Database Query Processing using open source DB
MongoDB/MySQL etc)

Software Requirements

1. Java / Python / R / Scala
2. Oracle, MySQL, MongoDB, Casandra, Hive

TOTAL :60 PERIODS

2402CA208

Professional Development Courses II

L	T	P	C
0	0	2	1

Prerequisite Basic understanding of communication skills, professional behavior, and quantitative aptitude fundamentals.

COURSE OBJECTIVES

1. To develop students' professional communication, interview, and teamwork skills for workplace readiness.
2. To enhance writing, listening, and speaking abilities required for effective professional interaction.
3. To strengthen problem-solving and analytical thinking through aptitude training and practice.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Enhance professional communication and interview skills through resume presentation, self-introduction, and mock interviews.
- CO2 Develop strong writing abilities in emails, essays, letters, and paragraphs while improving listening and speaking skills
- CO3 Apply quantitative aptitude skills to solve problems related to time, work, distance, profit & loss, and data interpretation.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	1	1	-	3	1	2	3	-	-
CO2	2	1	1	1	3	-	2	3	-	-
CO3	3	3	2	2	1	2	-	2	2	3
Avg	2.3	1.6	1.3	1.5	2.3	1.5	2	2.6	2	3

UNIT I Interview Essentials II 10 Hours

Interview Skills, developing professional behavior, Resume presentation, Self-introduction presentation, Interview etiquettes, Telephone etiquettes, Mock interview practice, Teamwork skills

UNIT II Writing Skills 10 Hours

E-mail writing, Essay writing, Letter writing, Paragraph writing, Listening and speaking skills

UNIT III Aptitudes 10 Hours

Time and Work - Time and Distance - Profit and Loss - Simple Interest and Compound Interest - Problem on Ages - Data Interpretations

TOTAL :30 PERIODS

REFERENCES:

1. Arun Sharma, _How to Prepare for Logical Reasoning for CAT_, 4th edition, McGraw Hills publication, 2017.
2. R S Agarwal, _A modern approach to Logical reasoning_, revised edition, S.Chand publication, 2017.
3. B.S. Sijwalii and Indu Sijwali, —A New Approach to REASONING Verbal & Non-Verbal, 2nd edition, Arihant publication, 2014
4. Objective General English by SP Bakshi
5. English Grammar in use by Murphy.

BRIDGE COURSES

Bridge Course (For the M.C.A students admitted under non-computer-science background category)						
Course Code	Course Name	L	T	P	C	Category
SEMESTER II						
2401CA003	Mathematical Foundations of Computer Applications	3	0	0	3	FC
2401CA004	Database Management Systems	3	0	2	4	PC
TOTAL		6	0	2	7	

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2401CA003	MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE	L	T	P	C
Prerequisite	Mathematical Foundations of Computer Science	3	0	0	3

COURSE OBJECTIVES

1. To introduce Mathematical Logic and their rules for validating arguments and programmes.
2. To introduce counting principles for solving combinatorial problems.
3. To give exposure to Graph models and their utility in connectivity problems.
4. To introduce abstract notion of Algebraic structures for studying cryptographic and its related areas.
5. To introduce Boolean algebra as a special algebraic structure for understanding logical circuit problems.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Apply Mathematical Logic to validate logical arguments and programmes
- CO2 Apply combinatorial counting principles to solve application problems.
- CO3 Apply graph model and graph techniques for solving network other connectivity related problems.
- CO4 Apply algebraic ideas in developing cryptograph techniques for solving network security problems.
- CO5 Apply Boolean laws in developing and simplifying logical circuits

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	3	1	-	2	2	3	2
CO2	3	3	2	2	1	-	-	2	3	3
CO3	3	3	3	3	2	1	-	2	3	3
CO4	3	3	3	3	1	1	3	2	3	3
CO5	3	2	3	3	1	-	2	2	3	2
Avg	3	2.6	2.6	2.8	1.2	1	2.3	2	3	2.6

UNIT I LOGIC AND PROOFS 9 Hours
 Propositional Logic – Propositional Equivalences – Predicates and Quantifiers – Nested Quantifiers – Rules of Inference – Introduction to Proofs – Proof Methods and Strategy.

UNIT II COMBINATORICS 9 Hours
 Mathematical Induction – Strong Induction and Well Ordering – The Basics of Counting - The Pigeonhole Principle – Permutations and Combinations – Recurrence Relations Solving Linear Recurrence Relations Using Generating Functions – Inclusion – Exclusion – Principle and Its Applications

UNIT III GRAPHS 9 Hours
 Graphs and Graph Models – Graph Terminology and Special Types of Graphs – Matrix Representation of Graphs and Graph Isomorphism – Connectivity – Euler and Hamilton Paths.

UNIT IV ALGEBRAIC STRUCTURES 9 Hours
 Groups – Subgroups – Homomorphisms – Normal Subgroup and Coset – Lagrange's Theorem – Definitions and Examples of Rings and Fields.

UNIT V LATTICES AND BOOLEAN ALGEBRA 9 Hours
 Partial Ordering – Posets – Lattices as Posets – Properties of Lattices – Lattices as Algebraic Systems – Sub Lattices – Direct Product And Homomorphism – Some Special Lattices – Boolean Algebra

TOTAL :45 PERIODS

REFERENCES:

1. Kenneth H. Rosen, "Discrete Mathematics and its Applications", Tata McGraw Hill Pub. Co. Ltd., Seventh Edition, Special Indian Edition, New Delhi, 2011
2. Tremblay J.P. and Manohar R, "Discrete Mathematical Structures with Applications to Computer Science", Tata McGraw Hill Pub. Co. Ltd, 30th Reprint, New Delhi, 2011.
3. Ralph. P. Grimaldi, "Discrete and Combinatorial Mathematics: An Applied Introduction", Pearson Education, 3rd Edition, New Delhi, 2014.
4. Thomas Koshy, "Discrete Mathematics with Applications", 2nd Edition, Elsevier Publications, Boston, 2006.
5. Seymour Lipschutz and Mark Lipson, "Discrete Mathematics", Schaum's Outlines, Tata McGrawHill Pub. Co. Ltd., Third Edition, New Delhi, 2013

2401CA004

DATABASE MANAGEMENT SYSTEMS

L	T	P	C
3	0	2	4

Prerequisite Basic knowledge of programming and data structures

COURSE OBJECTIVES

1. To understand the fundamentals of data models and conceptualize and depict a database system using ER diagram.
2. To make a study of SQL and relational database design.
3. To know about data storage techniques and query processing.
4. To impart knowledge in transaction processing, concurrency control techniques and recovery procedures.
5. To understand the fundamentals of data models and conceptualize and depict a database system using ER diagram.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Understand the basic concepts of the database and data models.
- CO2 Design a database using ER diagrams and map ER into Relations and normalize the relations
- CO3 Acquire the knowledge of query evaluation to monitor the performance of the DBMS
- CO4 Develop a simple database applications using normalization
- CO5 Acquire the knowledge about different special purpose databases and to critique how they differ from traditional database systems.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	1	-	-	2	3	2
CO2	3	3	3	3	2	1	-	2	3	2
CO3	3	3	3	3	1	1	-	2	3	3
CO4	3	2	3	3	2	1	-	2	3	2
CO5	3	2	2	2	1	-	-	2	3	3
Avg	3	2	2	2	1	1	-	2	3	2

UNIT I INTRODUCTION

9 Hours

File systems versus Database systems – Data Models – DBMS Architecture – Data Independence – Data Modeling using Entity – Relationship Model – E-R Modeling.

UNIT II RELATIONAL MODEL AND QUERY EVALUATION

9 Hours

Relational Model Concepts – Relational Algebra – SQL – Basic Queries – Complex SQL Queries – Views – Constraints

UNIT III DATABASE DESIGN & APPLICATION DEVELOPMENT

9 Hours

Functional Dependencies – Non-loss Decomposition – First, Second, Third Normal Forms, Dependency Preservation – Boyce/Codd Normal Form – Multi-valued Dependencies and Fourth Normal Form – Join Dependencies and Fifth Normal Form.

UNIT IV TRANSACTION PROCESSING

9 Hours

Query Processing-Transaction Processing – Properties of Transactions - Serializability – Transaction support in SQL - Locking Techniques – Validation Techniques – Recovery concepts – Shadow paging – Log Based Recovery.

UNIT V FILES AND INDEXING

9 Hours

File operations – Hashing Techniques – Indexing – Single level and Multi-level Indexes – B+ tree – Static Hashing

LIST OF EXPERIMENTS

Data Definition Commands to create, describe, alter, rename, drop and truncate the tables

1. Data Manipulation Commands for inserting, deleting, updating and retrieving in Tables
2. Transaction Control Language Commands like Commit, Rollback and Save Point
3. Illustrate the statements to create index and drop index
4. Perform database querying using simple query, nested query, subquery and join operations

5. Create a PL/SQL block to implement implicit and explicit cursors
6. Create a PL/SQL block to implement procedures and functions
7. Create a PL/SQL block to execute triggers
8. Execute a procedure which handles exception using PL/SQL
9. Create an embedded PL/SQL block to connect with any host language like 'C'

TOTAL :45 PERIODS

REFERENCES:

1. Abraham Silberschatz, Henry F.Korth and S.Sundarshan "Database System Concepts", Seventh Edition, McGraw Hill, 2017.
2. Ramez Elmasri and Shamkant Navathe, "Fundamentals of Database Systems", Seventh Edition, Pearson Education Delhi, 2017
3. Raghu Ramakrishnan, —Database Management Systems, Fourth Edition, McGraw Hill College Publications, 2015.
4. Lee Chao, "Database Development and Management", Auerbach Publications, 1st edition, 2010
5. Carlos Coronel, Peter Rob, and Stephen Morris, "Database Principles Fundamentals of Design, Implementation, and Management –10th Edition", Course Technology, Cengage Learning, 2013
6. C.J. Date, "An Introduction to Database Systems", Eighth Edition, Pearson Education Delhi, 2003

LIST OF AUDIT COURSES (AC) Registration for any of these courses is optional to students						
Course Code	Course Name	L	T	P	C	Category
2401AU004	Constitution of India	2	0	0	0	AC

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2401AU004	CONSTITUTION OF INDIA	L	T	P	C
Prerequisite	Basic understanding of Indian history and civics at school level.	2	0	0	0

COURSE OBJECTIVES

1. Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective
2. To address the growth of Indian opinion regarding modern Indian intellectuals' constitutional
3. Role and entitlement to civil and economic rights as well as the emergence nation hood in the early Indian nationalism
4. To address the role of socialism in India after the commencement of the Bolshevik evolution in 1917 and its impact on the initial drafting of the Indian Constitution

COURSE OUTCOMES

- CO1 Understand the history and drafting of the Indian Constitution.
 CO2 Explain the Preamble and key features of the Constitution.
 CO3 Describe Fundamental Rights, Duties, and Directive Principles.
 CO4 Understand the structure and functions of Parliament, Executive, and Judiciary.
 CO5 Explain local governance systems like Municipalities and Panchayati Raj.
 CO6 Understand the role of the Election Commission and welfare bodies for SC/ST/OBC/Women.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	-	1	1	-	2	1	3	2	-	-
CO2	-	1	1	-	2	1	3	2	-	-
CO3	-	2	1	-	2	2	3	2	-	-
CO4	-	1	1	-	2	1	3	2	-	-
CO5	-	2	1	-	2	2	3	2	-	-
CO6	-	1	1	-	2	1	3	3	-	-
Avg	-	1.3	1		2	1.3	3	2.1	-	-

UNIT I	HISTORY OF MAKING OF THE INDIAN CONSTITUTION	3 Hours
History, Drafting Committee, (Composition & Working)		
UNIT II	PHILOSOPHY OF THE INDIAN CONSTITUTION	3 Hours
Preamble, Salient Features		
UNIT III	CONTOURS OF CONSTITUTIONAL RIGHTS AND DUTIES	6 Hours
Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties.		
UNIT IV	ORGANS OF GOVERNANCE	6 Hours
Parliament, Composition, Qualifications and Disqualifications, Powers and Functions, Executive, President, Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions		
UNIT V	LOCAL ADMINISTRATION	6 Hours
District's Administration head: Role and Importance Municipalities introduction mayor and role of Elected Representative, CEO, Municipal Corporation. Pachayati raj: Introduction, PRI: Zila, Pachayat. Elected officials and their roles, CEO Zila Pachayat: Position and role. Block level: Organizational Hierarchy(Different departments), Village level: Role of Elected and Appointed officials, Importance of grass root democracy.		
UNIT VI	ELECTION COMMISSION	6 Hours
Election Commission: Role and Functioning. Chief Election Commissioner and Election Commissioners - Institute and Bodies for the welfare of SC/ST/OBC and women.		

TOTAL :30 PERIODS

REFERENCES:

1. Kai Hwang, Geoffrey C. Fox & Jack J. Dongarra, "Distributed and Cloud Computing, From Parallel Processing to the Internet of Things", Morgan Kaufmann Publishers, First Edition, 2012
2. drew S. Tanenbaum & Maarten Van Systems - Principles Third Edition, Pearson, 2017.
3. Thomas Erl, Zaigham Mahood & Ricardo Puttini, Computing, Concept, Technology & Prentice Hall, Second Edition, 2013.
4. Richard Rodger, Tao of ISBN 9781617293146, Manning Publications, First Edition, December 2017
5. On Microservices with Spring Boot and Spring Cloud: Build and deploy microservices using spring cloud, Istio and Packt Publishing Ltd, First Edition, September 2019.

PROFESSIONAL ELECTIVE I						
Course Code	Course Name	L	T	P	C	Category
2403CA001	Research Methodology and IPR	3	0	0	3	PE
2403CA002	Software Project Management	3	0	0	3	PE
2403CA003	Artificial Intelligence	3	0	0	3	PE
2403CA004	Data Analytics	3	0	0	3	PE
2403CA005	Soft Computing Techniques	3	0	0	3	PE
2403CA006	Digital Image Processing	3	0	0	3	PE

2403CA001

RESEARCH METHODOLOGY AND IPR

L	T	P	C
3	0	0	3

Prerequisite

COURSE OBJECTIVES

1. To impart knowledge and skills required for research and IPR
2. To learn the basis of Problem formulation, analysis, and solutions
3. To study the tools required for ethical research
4. To know about Patent drafting and filing patents.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Formulate Research Problems
- CO2 Analysis the given research topic.
- CO3 Implement research ethics in their research studies.
- CO4 Use modern tools in their research.
- CO5 Apply IPR ethics in their career

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	3	2	-	-	-	-	2	2	1
CO2	2	3	2	-	-	-	-	2	2	2
CO3	-	2	-	-	-	-	3	2	-	-
CO4	-	-	2	3	-	-	-	2	3	3
CO5	-	-	-	-	-	2	3	2	-	-
Avg	2	2.6	2	3	-	2	3	2	2.3	2

UNIT I

RESEARCH PROBLEM FORMULATION

9 Hours

Meaning of research problem- Sources of research problem, criteria characteristics of a good research problem, errors in selecting a research problem, scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, necessary instrumentations

UNIT II

L LITERATURE REVIEW

9 Hours

Effective literature studies approaches, analysis – Presenting literature surveys – Adding References

UNIT III

TOOLS FOR PROFESSIONAL AND ETHICAL RESEARCH

9 Hours

Plagiarisms – Various formats for writing Research Papers – Latex Tools – Plagiarism Checking Tools – Research Paper Repositories Online – Reference Management Tools.

UNIT IV

INTRODUCTION TO INTELLECTUAL PROPERTY RIGHTS (IPR)

9 Hours

Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.

UNIT V

INTELLECTUAL PROPERTY RIGHTS (IPR)

9 Hours

Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications. New Developments in IPR: Administration of Patent System, IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.

TOTAL :45 PERIODS

1. Asimov, "Introduction to Design", Prentice Hall, 1962.
2. Halbert, "Resisting Intellectual Property", Taylor & Francis Ltd, 2007.
3. Mayall, "Industrial Design", McGraw Hill, 1992.
4. Ranjit Kumar, Second Edition, "Research Methodology: A Step by Step Guide for beginners" 2010
5. Online Resources for Research

2403CA002

SOFTWARE PROJECT MANAGEMENT

L	T	P	C
3	0	0	3

Prerequisite

COURSE OBJECTIVES

1. To know how to do project planning for the software process.
2. To learn the cost estimation techniques during the analysis of the project.
3. To understand the quality concepts for ensuring the functionality of the software

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Understand the activities during the project scheduling of any software application.
 - CO2 Learn the risk management activities and the resource allocation for the projects.
 - CO3 Apply the software estimation and recent quality standards for evaluation of the software projects
 - CO4 Acquire knowledge and skills needed for the construction of highly reliable software project
 - CO5 Create reliable, replicable cost estimation that links to the requirements of project planning and managing

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	1	2	2	-	2	3	-	1	1	-
CO2	1	2	2	-	2	3	2	1	1	-
CO3	2	2	3	2	1	3	2	1	2	1
CO4	2	3	3	2	2	2	2	2	3	2
CO5	2	2	2	1	2	3	2	1	2	2
Avg	1.6	2.2	2.4	1.6	1.8	2.8	2	1.2	1.8	1.3

UNIT I SOFTWARE PROJECT MANAGEMENT CONCEPTS 9 Hours

Introduction to Software Project Management: An Overview of Project Planning: Select Project, Identifying Project scope and objectives, infrastructure, project products and Characteristics. Estimate efforts, Identify activity risks, and allocate resources- Six Sigma, Software Quality: defining software quality, ISO9126, External Standards.

UNIT II SOFTWARE EVALUATION AND COSTING 9 Hours

Project Evaluation: Strategic Assessment, Technical Assessment, cost-benefit analysis, Cash flow forecasting, cost-benefit evaluation techniques, Risk Evaluation. Selection of Appropriate Project approach: Choosing technologies, choice of process models, structured methods.

UNIT III SOFTWARE ESTIMATION TECHNIQUES 9 Hours

Software Effort Estimation: Problems with over and under estimations, Basis of software Estimation, Software estimation techniques, expert Judgment, Estimating by analogy. Activity Planning: Project schedules, projects and activities, sequencing and scheduling Activities, Networks planning models, formulating a network model.

UNIT IV RISK MANAGEMENT 9 Hours

Risk Management: Nature of Risk, Managing Risk, Risk Identification and Analysis, Reducing the Risk. Resource Allocation: Scheduling resources, Critical Paths, Cost scheduling, Monitoring and Control: Creating Framework, cost monitoring, prioritizing monitoring.

UNIT V GLOBALIZATION ISSUES IN PROJECT MANAGEMENT 9 Hours

Globalization issues in project management: Evolution of globalization- challenges in building global teams-models for the execution of some effective management techniques for managing global teams. Impact of the internet on project management: Introduction– the effect of the internet on project management – managing projects for the internet – effect on project management activities. Comparison of project management software: dot Project, Launchpad, openProj. Case study: PRINCE2

REFERENCE

TOTAL :45 PERIODS

1. Bob Hughes, Mike Cotterell & Rajib Mall "Software Project Management", McGraw-Hill Publications, 6th Edition 2017.
2. Ian Sommerville, "Software Engineering", 10th Edition, Pearson Education, 2017.
3. Robert T. Futrell, "Quality Software Project Management", Pearson Education India, 2008.

4. Gopalaswamy Ramesh, "Managing Global Software Projects: How to Lead Geographically Distributed
5. Teams, Manage Processes and Use Quality Models", McGraw Hill Education, 2017.
- Richard H. Thayer "Software Engineering Project Management", 2nd Edition, Wiley, 2006.
6. S.A. Kelkar, "Software Project Management" PHI, New Delhi, Third Edition, 2013

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2403CA003

ARTIFICIAL INTELLIGENCE

L	T	P	C
3	0	0	3

Prerequisite

COURSE OBJECTIVES

1. To understand the various characteristics and search strategies of Intelligent agents
2. To learn to represent knowledge in solving AI Problems
3. To understand different software agents design methods and its applications

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Explain the fundamentals, evolution, branches, and intelligent agent models of Artificial Intelligence.
- CO2 Apply appropriate problem-solving and search techniques to solve AI problems.
- CO3 Represent knowledge and reasoning using first-order predicate logic and logic programming techniques.
- CO4 Design and implement software agents with communication, negotiation, and reasoning capabilities.
- CO5 Develop AI-based applications using Natural Language Processing, Reinforcement Learning, and Intelligent Systems.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	-	2	-	-	-	2	2	2
CO2	3	3	2	3	-	-	-	2	3	2
CO3	3	3	2	2	-	-	-	1	2	2
CO4	2	3	3	3	-	-	-	2	3	3
CO5	2	2	3	3	-	-	-	2	3	3
Avg	2.4	2.6	2	2.6	-	-	-	1.8	2.6	2.4

UNIT I

INTRODUCTION TO AI

9 Hours

Introduction–Definition - AI Applications Future of Artificial Intelligence – Branches of AI - Evolution of AI - Intelligent Agents -Characteristics of Intelligent Agents–Typical Intelligent Agents – Agents And Environments- Good Behavior: The Concept of Rationality - The Nature of Environments - The Structure of Agents –Types of Structure Agent-Problem Solving Approach to Typical AI problems.

UNIT II

PROBLEM SOLVING METHODS

9 Hours

Problem solving Methods - Search Strategies- Uninformed - Informed - Heuristics - Local Search Algorithms and Optimization Problems - Local Search in Continuous Spaces - Online Search Agents and Unknown Environments -Searching with Partial Observations - Constraint Satisfaction Problems – Constraint Propagation - Backtracking Search - Game Playing - Optimal Decisions in Games – Alpha - Beta Pruning - Stochastic Games

UNIT III

KNOWLEDGE REPRESENTATION

9 Hours

Logical Agents -First Order Predicate Logic – Syntax and Semantics of First-Order Logic - Prolog Programming – Unification – Forward Chaining- Backward Chaining – Resolution – Knowledge Representation - Ontological Engineering-Categories and Objects – Events - Mental Events and Mental Objects - Reasoning Systems for Categories -Reasoning with Default Information- The Internet Shopping World

UNIT IV

SOFTWARE AGENTS

9 Hours

Architecture for Intelligent Agents – Agent communication – Negotiation and Bargaining –Argumentation among Agents – Trust and Reputation in Multi-agent systems– Prolog In Artificial Intelligence-Defining Prolog - Syntax and Meaning of Prolog Programs - Lists, Operators, Arithmetic - Operations on Data Structures - case Studies.

UNIT V

APPLICATIONS OF AI

9 Hours

AI applications – Natural Language Processing -Language Models – Text Classification-Information Retrieval- Information Extraction – Syntactic Analysis - Machine Translation – Speech Recognition - Reinforcement learning – Passive Reinforcement Learning - Active Reinforcement Learning - Robot – Hardware –Perception- Image Formation - Early Image-Processing Operations – Planning – Moving– Case Studies.

REFERENCE

TOTAL :45 PERIODS

1. S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach", Prentice Hall, Third Edition, 2009.
2. I. Bratko, —Prolog: Programming for Artificial Intelligence, Fourth edition, Addison-Wesley Educational Publishers Inc., 2011.
3. M. Tim Jones, —Artificial Intelligence: A Systems Approach(Computer Science), Jones and Bartlett Publishers, Inc.; First Edition, 2008
4. Gerhard Weiss, —Multi Agent Systems, Second Edition, MIT Press, 2013. 5. David L. Poole and Alan K. Mackworth, —Artificial Intelligence: Foundations of Computational Agents, Cambridge University Press, 2010.
5. <https://www.javatpoint.com/artificial-intelligence-tutorial>
6. https://www.tutorialspoint.com/artificial_intelligence/index.htm
7. <https://intellipaat.com/blog/tutorial/artificial-intelligence-tutorial/>

2403CA004

DATA ANALYTICS

Prerequisite Big Data Technologies

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

1. To understand fundamentals of BigData and Hadoop
2. To learn about file system configuration in HADOOP
3. To learn MapReduce concept of Hadoop in executing Task
4. To learn the Queue Processing and stream processing of Data
5. To learn about Hadoop Frameworks

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Apply Hadoop for analyzing Big Volume of Data
- CO2 Implement Big Data Activities using Hadoop
- CO3 Design of Algorithms to solve Data Intensive Problems using Map Reduce Paradigm
- CO4 Design and Implementation of Stream processing using kafka
- CO5 Implement Big Data Activities using Hadoop Framework

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	3	-	-	-	2	3	3
CO2	3	3	2	3	-	-	-	2	3	3
CO3	2	3	3	3	1	-	-	2	3	3
CO4	2	2	3	3	1	-	-	2	3	3
CO5	3	2	2	3	1	-	-	2	3	3
Avg	2.6	2.4	2.2	3	1	-	-	2	3	3

UNIT I INTRODUCTION TO BIGDATA AND HADOOP

9 Hours

Types of Digital Data - Introduction to Big Data -Challenges of conventional systems - Web data – Evolution of Analytic scalability - Analytic Processes and Tools -Analysis vs Reporting -History of Hadoop- Apache Hadoop - Analyzing Data with Hadoop- Hadoop Streaming

UNIT II HDFS & HADOOP I/O

9 Hours

Hadoop Distributed File System :The Design of HDFS- HDFS Concepts- The Command-Line Interface- Hadoop File Systems- Data Flow-Parallel Copying with distcp-Hadoop Archives- Hadoop I/O:Data Integrity- Compression-Serialization

UNIT III MAPREDUCE

9 Hours

Analyzing the Data with Hadoop- Hadoop Pipes- MapReduce Types - Input Formats- Output Formats- MapReduce Features - MapReduce Works - Anatomy of a MapReduce Job Run –Failures -Job Scheduling- Shuffleand Sort -Task Execution

UNIT IV QUEUEING AND STREAM PROCESSING SYSTEMS

9 Hours

Queueing: Queueing systems, Introduction to kafka, producer consumer, brokers, types of queues - Single consumer, multi consumer queue servers. Streaming systems: Stream processing–queues and workers- micro batch streaming processing Introduction to kafka streaming processing API

UNIT V HADOOP FRAMEWORKS

9 Hours

Pig: Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators. Hive : Hive Shell, Hive Services, Hive Meta store, Comparison with Traditional Databases, Hive QL, Tables, Querying Data

REFERENCE

TOTAL :45 PERIODS

1.Bill Franks, Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with advanced analytics,JohnWiley&sons,2012.

2.Michael Berthold,DavidJ. Hand, Intelligent DataAnalysis, Springer,2007

3.Tom White,Hadoop:TheDefinitiveGuide,O'Reilly, 2009

4.Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles ,DavidCorigan,"HarnessthePowerBigDataTheIBMBigDataPlatform",TataMcGrawHillPublications,2012.

5.Kafka: The Definitive Guide- Real-Time Data and Stream Processing at Scale,by

GwenShapira,Neha Narkhede,ToddPalino

2403CA005

SOFT COMPUTING TECHNIQUES

L	T	P	C
3	0	0	3

Prerequisite Basic knowledge of Mathematics.

COURSE OBJECTIVES

1. To gain knowledge of soft computing theories and its fundamentals
2. To design a soft computing system required to address a computational task.
3. To learn and apply artificial neural networks, fuzzy sets and fuzzy logic and genetic algorithms in problem solving and use of heuristics based on human experience.
4. To introduce the ideas of fuzzy sets, fuzzy logic and to become familiar with neural networks that can learn from available examples and generalize to form appropriate rules for inferencing systems.
5. To familiarize with genetic algorithms and other random search procedures while seeking global optimum in self-learning situations

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Identify and describe soft computing techniques and their roles in building intelligent Machines
 - CO2 Recognize the feasibility of applying soft computing methodology for a particular problem
 - CO3 Apply fuzzy logic and reasoning to handle uncertainty and solve engineering problems.
 - CO4 Apply genetic algorithms to optimization problems
 - CO5 Design neural networks to pattern classification and regression problems using a soft computing approach.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	-	-	-	2	2	1
CO2	2	3	2	2	-	-	-	2	2	2
CO3	3	3	3	2	-	-	-	2	2	2
CO4	3	2	3	2	-	-	-	2	3	2
CO5	3	2	3	3	-	-	-	2	3	3
Avg	2.8	2.4	2.6	2.2	-	-	-	2	2.4	2

UNIT I FUZZY COMPUTING

9 Hours

Basic Concepts of Fuzzy Logic, Fuzzy Sets and Crisp Sets, Fuzzy Set Theory and Operations, Properties of Fuzzy Sets, Fuzzy and Crisp Relations, Fuzzy to Crisp Conversion Membership Functions, Interference in Fuzzy Logic, Fuzzy If – Then Rules, Fuzzy Implications and Fuzzy Algorithms, Fuzzification and Defuzzification, Fuzzy Controller, Industrial Applications.

UNIT II FUNDAMENTALS OF NEURAL NETWORKS

9 Hours

Neuron, Nerve Structure and Synapse, Artificial Neuron and its Model, Activation Functions, Neural Network Architecture: Single Layer and Multilayer Feed Forward Networks, Recurrent Networks. Various Learning Techniques; Perception and Convergence Rule, Auto-Associative and Hetero-Associative Memory

UNIT III BACKPROPAGATION NETWORKS

9 Hours

Back Propagation Networks Architecture: Perceptron Model, Solution, Single Layer Artificial Neural Network, Multilayer Perceptron Model; Back Propagation Learning Methods, Effect of Learning Rule Co – Efficient ; Back Propagation Algorithm, Factors Affecting Backpropagation Training, Applications

UNIT IV COMPETITIVE NEURAL NETWORKS

9 Hours

Kohonen's Self Organizing Map – SOM Architecture, learning procedure – Application; Learning Vector Quantization – learning by LVQ; Adaptive Resonance Theory – Learning procedure – Applications.

UNIT V GENETIC ALGORITHM

9 Hours

Basic Concepts, Working Principle, Procedures of GA Flow Chart of GA, Genetic Representations, (Encoding) Initialization and Selection, Genetic Operators, Mutation, Generational Cycle, Applications

REFERENCE

TOTAL :45 PERIODS

1. J.S.R. Jang, C.T. Sun and E. Mizutani, "Neuro-Fuzzy and Soft Computing", Pearson Education, 2004.
2. Sekaran and G.A. Vijayalakshmi Pai, "Neural Networks, Fuzzy Systems and Evolutionary Algorithms: Synthesis and Applications", PHI Learning, 2nd Edition, 2017.
3. S.N. Sivanandam, S.N. Deepa, "Principles of Soft Computing", Third Edition, Wiley, 2018.
4. Simon Haykin, "Neural Networks and Learning Machines", Pearson, 3rd Edition, 2009.
5. Timothy Ross, "Fuzzy Logic with Engineering Applications", Wiley Publications, 4th Edition 2016.

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DIGITAL IMAGE PROCESSING

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Prerequisite Basic knowledge of Mathematics.

COURSE OBJECTIVES

1. Learn digital image fundamentals.
2. Be exposed to simple image processing techniques
3. Learn to represent image enhancement in the spatial and frequency domain.
4. Be familiar with image segmentation and compression techniques

COURSE OUTCOMES

- Upon completion of the course, the students will be able to
- CO1 digitize the input image using appropriate sampling and quantizing techniques
- CO2 Transform the input images to various domains and classify the images
- CO3 enhance the images using spatial domain and frequency domain for better visual representation
- CO4 To extract the features of a image by applying Morphological Image Processing techniques.
- CO5 Analyze the different image compression techniques and its significance

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	2	1	-	-	1	3	2
CO2	3	3	2	3	1	-	-	1	3	3
CO3	3	2	2	3	1	-	-	1	3	2
CO4	2	3	2	3	1	-	-	1	3	3
CO5	3	2	2	3	1	1	-	1	3	3
Avg	2.8	2.4	1.8	2.8	1	1	-	1	3	2.6

UNIT I DIGITAL IMAGE FUNDAMENTALS

9 Hours

Elements of visual perception, Image Acquisition Systems, Sampling and Quantization, Image Formation, Image Geometry, Different types of digital images. Relationship between pixels, Basic concepts of distance transform, Color Image fundamentals-RGB-HIS Models, Different color models-conversion.

UNIT II IMAGE TRANSFORMS

9 Hours

1D Discrete Fourier Transform (DFT), 2D transforms – DFT, Discrete Cosine Transform, Walsh and PCA

UNIT III IMAGE ENHANCEMENT

9 Hours

Gray Level transformations, Histogram Equalization, Spatial Domain: Basics of Spatial Filtering: smoothing and sharpening spatial filters. Frequency domain: smoothing and sharpening frequency domain filters, Ideal, Gaussian filters.

UNIT IV IMAGE SEGMENTATION AND FEATURE EXTRACTION

9 Hours

Segmentation: Point detection, line detection, edge detection, Region based segmentation, Region Splitting and Merging Technique. Thresholding Techniques: multilevel thresholding, optimal thresholding using Bayesian classification. Feature Extraction: GLCM, Hough Transform, Morphological operation

UNIT V IMAGE COMPRESSION

9 Hours

Lossy and lossless compression schemes, prediction based compression schemes, sub-band encoding schemes, JPEG compression standard, Fractal compression scheme, Wavelet compression scheme

REFERENCE

TOTAL :45 PERIODS

1. Rafael C.Gonzalez and Richard E.Woods, "Digital Image Processing", 4th Edition, Pearson Education, New Delhi, 2018
2. Jain Anil K., "Fundamentals of Digital Image Processing", 1st Edition, Prentice Hall of India, New Delhi, 2002.
3. Kenneth R.Castleman, "Digital Image Processing", 1st Edition, Prentice Hall of India, New Delhi, 2006.
4. John C.Russ, "The Image Processing Handbook", 5th Edition, Prentice Hall, New Jersey, 2002.
5. Willliam K Pratt, "Digital Image Processing", 3rd Edition, John Willey, 2002.
6. Dr.S.Sridhar, Digital Image Processing, Second Edition, Oxford University Press, 2016.